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APPLIED RESEARCH

Peruvian Presidential Debates in the Elections of 2021 in Twitter/X: A Sentiment Analysis Approach

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ABSTRACT Over the years, technology has rapidly evolved, with social networks and media emerging as prominent examples. Social media facilitates the creation and exchange of content, while social networks connect individuals and organizations based on shared interests or values. Among these platforms, Twitter (now X) has gained significant attention and has been a valuable tool for monitoring reputation and brands, including political discourse. With the widespread use of social networks, the demand for advanced textual data analysis techniques, particularly sentiment analysis, has grown. Sentiment analysis, or polarity classification, aims to gauge the positivity or negativity of textual content, providing information on public opinion dynamics. In the context of the Peruvian presidential elections of 2021, characterized by political uncertainty exacerbated by the COVID-19 pandemic, social media platforms, especially Twitter/X, emerged as vital arenas for political discourse and participation. The election featured various candidates representing a wide spectrum of ideologies, reflecting the complexities of Peruvian society. Against this backdrop, this research uses Twitter/X data to analyze public sentiment toward presidential candidates and election-related topics. Through state-of-the-art sentiment analysis algorithms, this study categorizes tweets into positive, negative, or neutral sentiments, revealing key trends and themes driving public opinion on Twitter/X during the election period. By examining sentiment dynamics in response to major events and comparing Twitter/X sentiment trends with official voting data, the research aims to assess the predictive power of social media sentiment analysis in forecasting electoral outcomes. Therefore, this research contributes to understanding the role of social media in shaping political discourse and public opinion during elections. By conducting sentiment analysis on Twitter data, valuable insights are offered to policymakers, political analysts, and researchers who want to leverage social media to understand and forecast electoral trends.

INDEX TERMS Machine learning, sentiment analysis, text mining, social network, Twitter/X, Peru, presidential elections.

I. INTRODUCTION

Over the years, technology has shown an accelerated evolution; social networks and media are two remarkable examples. Meanwhile, social media is defined as a group of applications built on the Internet to allow the creation and exchange of content [1]; a social network is a structure that aims to connect people and organizations based on a common interest or value [2]. One of these social networks is Twitter (currently known as ''X'', hereinafter defined as Twitter/X). With more than 400 million users and one million messages per day, Twitter/X has quickly become a valuable

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asset for organizations to invigilate their reputation and brands by extracting and analyzing the sentiment contained in the public-published tweets about their products, market services, and even their competitors [1]. Among these organizations are political entities that share their thoughts through this platform, providing the opportunity to interact with them and making Twitter/X a great place to collect meaningful political opinions and insights.

Along with the recognition of social networks worldwide, the need for textual data analysis techniques to produce meaningful information has become fundamental. Sentiment analysis, or statement polarity classification, is concerned with determining the relative positivity or negativity of a document, web page, or selection of text [1]. Several studies have shown that sentiment analysis is especially suitable in the election season.

For example, in 2016, it was used to predict the presidential elections of the United States of America [3], [4], [5], [6]. Similarly, in 2020, these techniques were implemented to predict electoral outcomes in the same country [7]. Additionally, in India, various research efforts focused on the 2019 presidential elections, the 2019 Prime Ministerial elections, and the 2017 Punjab state legislative assembly elections [8], [9], [10]. This methodology was also used to assess the Philippine presidential elections in 2022 [11] and the 2017 Jakarta gubernatorial elections in Indonesia [12]. Furthermore, it was applied to the Spanish presidential elections of 2019. In many of these studies, the efficacy of this approach in predicting actual results after the official count of votes has been demonstrated [7], [9], [10], [12], [13].

In the Peruvian context, political uncertainty has been a common occurrence for decades. However, the health crisis induced by COVID-19 exacerbated this situation. In the 2021 Peruvian presidential elections, the initial phase saw significant support for a range of candidates, reflecting the country's diverse political spectrum. Among the notable candidates in the early stages were Yonhy Lescano of Acción Popular, known for his centrist policies, and Verónika Mendoza of Juntos por el Perú, representing the left-wing segment. However, as the election progressed, the finalists emerged as Pedro Castillo from Peru Libre, a far left party advocating for major structural changes in the Peruvian political and economic system, and Keiko Fujimori from Fuerza Popular, a right-wing party with a focus on conservative policies and economic liberalism.

This dichotomy between Castillo and Fujimori's political alignments signified a profound division in the electorate, highlighting the stark ideological contrasts within Peruvian society and the challenges in achieving a consensus across such diverse political views. Consequently, social media platforms, particularly due to the restrictions imposed by the pandemic, became vital forums for the exchange of ideas. Twitter/X and other social networks thus became the primary channels for political discourse and electoral engagement for a significant portion of the electorate.

Building upon this scenario where social media platforms have become central to political discourse, the current research uses these digital spaces as a rich source of data. Considering that social networks serve as platforms where individuals freely share their views, a wide array of distinct perspectives can be gathered and examined. This analysis aims to uncover the underlying sentiments, offering insights into human behavior and preferences that are crucial in decision-making and predictive tasks [14].

Therefore, the objective of this study is to meticulously analyze the corpus of tweets related to the Peruvian presidential elections of 2021. This analysis specifically focuses on tweets about the candidates, including those related to debates and weekly published polls. The ultimate goal is to use these insights to predict the outcome of the elections, demonstrating the potential of sentiment analysis in contemporary political contexts.

The main contribution of this study presents an innovative method by combining sophisticated sentiment analysis techniques with social media data to accurately predict election results. It surpasses the existing literature by thoroughly assessing the effectiveness of different feature extraction models and machine learning algorithms in representing public sentiment toward political candidates. Furthermore, our research presents an introduction to the use of bigram analysis to identify subtle language patterns, which may provide a deeper understanding of voter preferences and behaviors.

The structure of this paper is organized as follows. Section II covers a review of the literature pertinent to this research. This is followed by a detailed description of the methodology adopted. Subsequent sections present the results and engage in a discussion contextualized within the framework of the existing literature. Finally, the article concludes with a summary of the research findings and implications.

II. RELATED WORKS

Currently, Twitter/X users post their views in 280-character text posts, with premium users having a limit of 25,000 characters. However, everyday interactions enable large volumes of textual data, resulting in a potentially large corpus. This abundance of data not only helps commercial entities in deciphering consumer preferences [1], but also provides information on the political sphere, especially to gauge voter sentiments toward candidates during election periods [12], [13].

Expanding the scope to a global scale, several studies have been conducted on presidential elections using similar digital data analysis techniques.

In the Spanish elections of 2019, in [15] the authors constructed a novel data set due to the lack of freely available data employing sentiment analysis tools and autoencoders for a temporal sentiment assessment, reveling in non-parallel sentiment profiles between political parties. In [16] five Machine Learning algorithms (Logistic Regression, Linear

Discriminant Analysis, K-Nearest Neighbors, Support Vector Machines and Random Forest) underwent evaluation to predict the results of Spanish elections and the candidate's voting share. In Indonesia's 2019 presidential elections [8], a data set was gathered before and after the elections, with sentiment analysis conducted using machine learning algorithms such as Support Vector Machines, Logistic Regression, and Multinomial Naive Bayes, supplemented by Deep Learning techniques for enhanced accuracy. In line with previous work in [17], Long-Short Term Memory and Bidirectional Encoder Representations from Transformers techniques from the field of Deep Learning were used to learn the opinion of the electorate concerning the top three presidential candidates in the 2023 Nigerian elections.

Focusing on the 2022 Philippine presidential elections [11], the investigation went beyond data collection and pre-processing, exploring the predominant sentiments expressed by the electorate and their impact on the election outcome. The findings of this study showed the polarization of public opinion, highlighting the significance of negative sentiments regardless of the candidate's position. Similarly, [18] included eight emotions in a sentiment analysis framework based on Twitter/X data to show the effects of electorate emotions on the Brazilian presidential election of 2020. The results showed a strong correlation between a prediction provided by an emotion-based XGBoost and Random Forest models and the election results.

Furthermore, the work on India's Lok Sabha 2019 elections [9] implemented a transfer learning approach to better understand the complexities of political discourse on social media, including the detection of sarcasm.

Lastly, these techniques have also been used in the presidential elections of the United States of America in 2016. In [3] and [6] they collected tweets that only mentioned one candidate and classified them by state. They then developed a sentiment analysis for the tweets of each candidate in which the words were assigned a score according to their sentiment, and when there was a form of negation, the score of the next word was changed to the other sign. Then, to predict the winner in each state, the percentage of positive messages was compared. In [3], when comparing the prediction with the electoral result, a correlation of 66.7% was found between Twitter/X sentiments and the true electoral winner. In [6], the authors were able to correctly classify 31 of the 50 states and achieved an accuracy of 62%.

More recently, [7] involved analyzing Twitter/X data before, during, and after the 2020 US presidential elections to measure public sentiment and compare it with election results. It employed TF-IDF, short for Term Frequency-Inverse Document Frequency, to extract features from tweets and the Naive Bayes classifier for sentiment analysis. The results were mostly in agreement with the actual election results, except for four notable deviations, indicating regional variations in sentiment. In [19], the authors used a multimodal Naive Bayes classifier to predict the 2020 US presidential election. The classifier operated on geolocated textual data from Tweeter/X before, during, and after the election day. Their results showed that geolocating tweets offers predictions aligned with the actual results of the US presidential election.

Expanding the scope of sentiment analysis in other political contexts, several studies have focused on elections beyond the presidential level. In India, the sentiment towards the prime ministerial candidates was analyzed [13], using Twitter/X's API to gather relevant data. This study used tools such as R and Rapid Miner for sentiment analysis, and its results, which favored one candidate over the other, were later validated by actual election results. Similarly, the 2017 gubernatorial elections in DKI Jakarta were subject to sentiment analysis [12]. This research utilized machine learning algorithms such as Multinomial Naive Bayes and Support Vector Machine for predictive analysis, complemented by descriptive tools like time-series graphs and word clouds. The study's findings underscored the accuracy of continuously analyzing public sentiment to understand election outcomes.

Another study focused on the Indian general elections in 2017 [10], studying public opinion about political parties such as AAP, BJP, and INC. The data collected by Twython were analyzed using the Naive Bayes algorithm. The analysis revealed a higher volume of negative tweets from AAP compared to other parties, a sentiment that was reflected in the actual election results. Reflecting the growing trend of employing sentiment analysis in diverse political landscapes, the research has also been extended to specific national contexts with significant results.

A prime example of this is the study of political sentiments in Peru, where a notable instance occurred in September 2020, during the presidential vacancy process involving Martín Vizcarra, the president at that time. The study by Saire and Zuñiga [20] serves as a key reference in the analysis of the impeachment process, which involves the collection and purification of tweets from Peru. Their methodology consisted of quantifying tweet volumes by region, identifying prevalent keywords, and tracking their daily variations. In addition, the study conducted a sentiment analysis, categorizing the tweets into positive and negative sentiments, and observing their fluctuation over time. Furthermore, trends in Google and YouTube searches were also analyzed to obtain more information, and they found that data from these platforms can be a useful resource for analyzing and detecting political events in specific locations.

Finally, in [4] the authors analyze the tweets of the different US electoral states together, which is more like the style of the Peruvian election. In this investigation, they determine the polarity and subjectivity measured for the collected tweets that help to understand the user's opinion of a particular candidate. In addition, a comparison is made between the candidates on the type of sentiment. It was inferred from the results that Hillary won due to the number of positive tweets, thus giving a better average score polarity than Bernie and

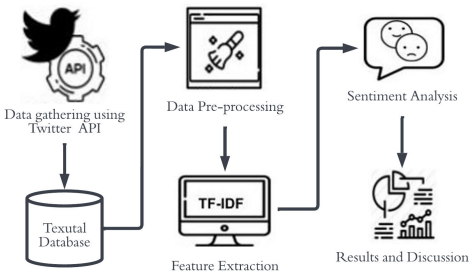


FIGURE 1. Framework for analysis of presidential elections using Twitter/X.

Trump. In the end, Clinton won the popular vote, but Trump the electoral vote.

In conclusion, the studies described above demonstrate the widespread application and efficacy of sentiment analysis techniques in deciphering electoral trends and voter sentiments in various political landscapes, including the United States, India, Indonesia, Spain, and Peru. Building upon this rich body of literature, the current research aims to contribute to this growing field by focusing specifically on the Peruvian presidential elections of 2021. Using sentiment analysis methodologies and exploiting the vast amounts of data generated on social media platforms such as Twitter/X, this study seeks to uncover valuable insights into the dynamics of public opinion surrounding the Peruvian electoral landscape. Through a meticulous analysis of Twitter/X data related to candidates, debates, and polling trends, the research aims to offer a nuanced understanding of the sentiments expressed by Peruvian voters during the election period. Therefore, this research represents a significant step forward in the application of sentiment analysis techniques in the Peruvian political context, providing policymakers, political analysts, and researchers with valuable resources to navigate the complexities of the country's electoral environment and anticipate social changes. Consequently, the findings of this study are poised to contribute to a deeper understanding of political discourse and decision-making processes, paving the way for more informed and data-driven approaches to studying and forecasting electoral outcomes.

III. METHODOLOGY

The methodology described in Figure 1 provides a roadmap for our analysis, beginning with the collection of tweets and moving through the data cleaning to prepare the data set. We then proceed to vectorization, enriching the data for sentiment analysis. This leads to the development of predictive models and the process culminates in the visualization of our results. All data used in this research are available in an online repository.[1]

[1]https://n9.cl/tweetsdata

A. RECOLLECTING THE DATA

The initial step of the investigation involved collecting data on the general opinions of Peruvians about the presidential candidates. This data was collected using the Twitter/X API during the 2021 presidential debates. Specifically, the two debates conducted by the Peruvian official authorities on 23 May and 30 May were analyzed, since presidential debates are crucial in shaping voters' opinions about candidates [21]. Tweets were collected approximately 30 minutes before the debates and up to 60 minutes after they concluded. This time frame effectively captures user opinions while the topic remains relevant to social media, as interactions tend to decrease and sentiments lose intensity over time [22]. Nonetheless, to capture the larger sentiment about the two candidates, we have also collected tweets corresponding primarily to the days near the debates, likewise, tweets from other days in May and June are also included.

To retrieve these tweets, three filters were used: location, language, and word. The tweet had to be made in Peru and Spanish. For word filtering, if we recall tweets about the polls, the name of the company that published the poll had to be mentioned. For each debate, the names of the candidates and the hashtags of the debate were also used as filters.

B. CLEANING THE DATA

After collecting the tweets, we ensured that each tweet mentioned only one candidate. This helped us to clearly identify the sentiment towards that candidate. To obtain a more accurate view of the overall sentiment of the election, we then selected only one tweet per user per week. This approach mirrors the real-world voting system, where each person receives only one vote, thus preventing the opinion of any single user from disproportionately influencing our sentiment analysis and avoiding fake trends generated by a user with many interactions. Furthermore, this method helps map the impact of the debates on user opinions more accurately by reflecting each person's vote only once. Then, the tweets were tokenized, all the words were returned in lowercase, and the stop words determined by the library NLTK were removed, except the word ''no'', since the absence of this adverb might change the sense of the statement. In addition, all numbers, hashtags, and tags were removed.

C. VECTORIZATION

For the vectorization and sentiment analysis steps, 10% of the data was manually labeled (approximately 6,900 tweets for Keiko and 5,100 for Pedro) with a positive sentiment (1) or negative sentiment (0). This labeled dataset was then divided into training and testing subsets (80/20 split). Each candidate was independently processed, each having its own vectorizer and sentiment classifier.

In this research, we used three types of feature extractors: Term Frequency-Inverse Document Frequency (TF-IDF)

[7], [8], [9], [11], [23], which evaluates the relevance of words by balancing their frequency in specific documents against their occurrence across all documents; Word2Vec (W2V) [24], a neural network model that translates words into vectors to represent semantic relationships; and Bag of Word Embeddings (BOWE) [23], [25], which extends W2V by aggregating word vectors to form a comprehensive representation of text. The integration of BOWE with W2V captures both the individual semantic meanings of words and their collective context in the text. These techniques enable the representation of textual information in one-dimensional arrays, which makes it a suitable input for machine learning algorithms and the prediction-making process.

Various configurations for each vectorizer were tested using the grid search hyperparameter optimization method. Each vectorizer configuration was evaluated based on its classification performance using a default configured SVM model [26], with the F1-score as the classification metric. For TF-IDF, 864 combinations were tested; for W2V, 384; and for BOWE, 192. Not all desired combinations for BOWE were tested due to resource and time constraints. The set of parameters and their values tested in each case are shown in Tables 1, 2, and 3 for each respective vectorizer. These experiments aimed to identify the optimal configurations for each vectorizer to improve classification performance, using 5-fold cross-validation for robust assessment.

The Figures 2 and 3 show the best three configurations for each vectorizer for Keiko Fujimori and Pedro Castillo, respectively. These configurations are measured by the F1-score, and the best configuration within them are highlighted in orange in the aforementioned figures. These graphs describes that for both candidates, the TF-IDF vectorizer yielded the best results, followed by W2V, with BOWE showing inferior performance compared to the others.

The best vectorizer configurations for each case are highlighted in bold in the corresponding tables. Bold indicates the best settings for both, red indicates the best for Pedro Castillo, and blue indicates the best configurations for Keiko Fujimori. Finally, in the next stage such best configurations are used along with machine learning (ML) models for the sentiment classification of each tweet.

TABLE 1. Tested parameters for TF-IDF.

Parameter	Tested Values
Max Features	500, 750, 1000, 1250
Ngram Range	(1, 1), (1, 2), (1, 3)
Max df	0.75, 0.85, 0.95, 1.0
Min df	1, 2, 5, 10
Norm	`"l1"`, **`"l2"`**

D. SENTIMENT ANALYSIS

The sentiment analysis framework used in this study is detailed in Figure 4 and was carried out using four ML models: Decision Tree (DT), which divides data into branches to make decisions based on feature values; Random Forest (RF), an ensemble of decision trees that improves accuracy and reduces overfitting; Support Vector Machine (SVM), which finds the optimal hyperplane to separate data points of

TABLE 2. Tested parameters for Word2Vec.

Parameter	Tested Values
Vector Size	**100**, 200, 300
Min Count	1, 2, 5, 7
Window	3, 5, 7, **10**
Sg	0, **1**
Alpha	0.01, 0.025, **0.05**
Epochs	5, 10, **20**

TABLE 3. Tested parameters for BOWE.

Parameter	Tested Values
Number of Clusters	50, 75, 95, 135, 150, **200**
Init	`"k-means++"`, **`"random"`**
Tolerance	1e-4, 1e-3

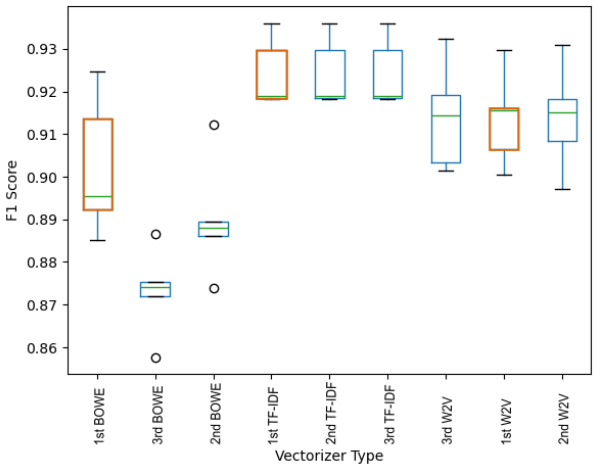


FIGURE 2. Top 3 configurations of three different vectorizers - Keiko Fujimori.

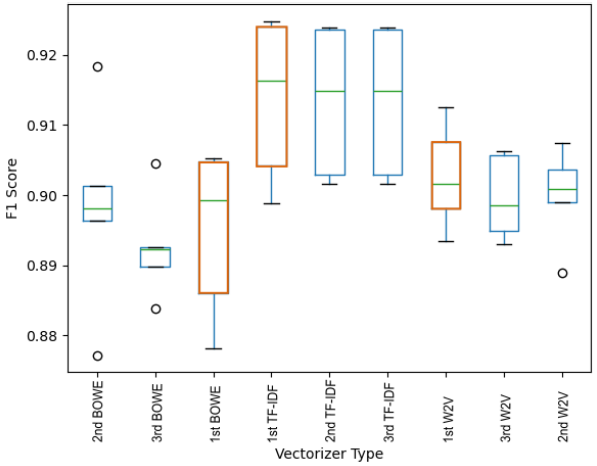


FIGURE 3. Top 3 configurations of three different vectorizers - Pedro Castillo.

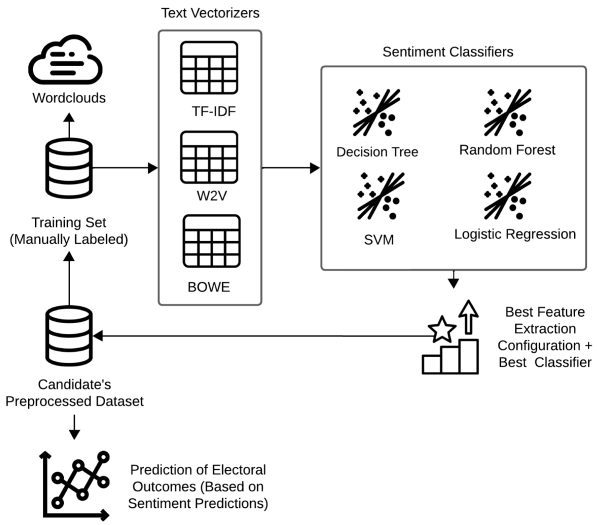


FIGURE 4. Overview of sentiment analysis workflow.

different classes; and Logistic Regression (LR), a statistical model that predicts the probability of a binary outcome [27].

As described in Figure 4, the core of sentiment analysis is based on the use of the best feature extractor to convert the tweets of each candidate into vectors; therefore, the process began by vectorizing the tweets related to each candidate using the aforementioned selected feature extraction models.

For the classification stage, two machine learning sentiment analysis models were constructed for each candidate: one for Pedro Castillo and the other for Keiko Fujimori. For each candidate, their three different sets of vectors (the best configuration of each of the three vectorization models) were tested. In addition, various parameter configurations for each classification model were evaluated through a grid search. The classification parameters tested are shown in Tables 4, 5, 6, and 7, respectively. In total, 2,304 different models of DT, 1,158 of RF, 778 of SVM, and 286 of LR were evaluated. The best parameter configuration for each classification model is highlighted in bold in its corresponding table. Bold indicates the best for both, red indicates the best for Pedro Castillo, and blue indicates the best for Keiko Fujimori.

TABLE 4. Tested parameters for decision tree.

Parameter	Tested Values
Criterion	`"gini"`, `"entropy"`
Splitter	`"best"`, **`"random"`**
Max Depth	10, 20, 30, `None`
Min Samples Split	2, 5, 10
Min Samples Leaf	**1**, 2, 4, 10
Class Weight	`None`, `"balanced"`

After the ML classifier training process was completed, the feature extraction model together with the classification model that yielded the highest F1-score for each candidate's data set was selected to process the feature vectors of the complete tweet data set, so the classification model was able

TABLE 5. Tested parameters for random forest.

Parameter	Tested Values
Number of Estimators	50, 100, 200, 500
Max Depth	20, 30, `None`
Min Samples Split	2, 5, 10
Min Samples Leaf	1, 2, 4, 10
Class Weight	`None`, **`"balanced"`**

TABLE 6. Tested parameters for SVM.

Parameter	Tested Values
C	0.001, 0.01, 0.1, 1, 10, 100, 1000
Kernel	`"linear"`, `"poly"`, **`"rbf"`**
Gamma	`"scale"`, `"auto"`, 0.001, 0.01, 0.1, 1, 10
Class Weight	**`None`**, `"balanced"`

TABLE 7. Tested parameters for logistic regression.

Parameter	Tested Values
C	0.001, 0.01, 0.1, 1, **10**, 100
Solver	`"newton-cg"`, `"lbfgs"`, `"liblinear"`
Penalty	`"l1"`, **`"l2"`**
Class Weight	**`None`**, `"balanced"`

to label the evaluation data set adequately with the remaining unlabeled tweets, to subsequently perform the prediction of the electoral results based on these sentiment classification results.

Finally, an exploratory analysis of manually labeled data was performed using a word cloud approach that focused on identifying the most common words within the comments (positive and negative) about each presidential candidate. This process was carried out to determine the characteristic words associated with positive and negative sentiments towards each candidate. It should be noted that the sentiment analysis was performed in the original language of the publications, that is, Spanish. Subsequently, the results were translated into English to facilitate greater understanding.

E. PREDICTED RESULTS

In examining presidential elections, this study adopts a methodology analogous to that utilized in previous research [3], [4], [5], [6], [7], [9], [11], [12], [20] to assess voting patterns. Specifically, we employ two distinct aggregation methods for Twitter/X data to predict election outcomes. This approach aims to understand the voting tendencies in Peru, as indicated by the sentiment expressed on Twitter/X during the four weeks of May 2021.

The first method adopted an additive approach, that is, for each candidate, it summed the number of positive tweets they received and added this to the number of negative tweets directed at their opponent, doing this calculation for each of the four weeks. This model was designed to provide an aggregate sentiment score, giving us an idea of the overall

positive support for a candidate while also considering the negativity faced by their rival.

The second method, on the other hand, took a differential approach. It calculated the proportion of negative tweets for each candidate and subtracted this from the proportion of their positive tweets, again performing this analysis weekly. This model was designed to offer a more nuanced view of the sentiment landscape, focusing on the net sentiment (positive minus negative) for each candidate.

By contrasting these two models, we aim to capture both the direct support for candidates (as seen in the first model) and the net sentiment balance (as shown in the second model), providing a comprehensive understanding of public sentiment during the election period.

F. VISUALIZATION

The two predictive aggregation methods described previously were visualized with line graphics that showed changes in sentiments about each candidate over time. In addition, four-word clouds are displayed, illustrating the most frequent words associated with both positive and negative viewpoints on each candidate. As a foundational effort to comprehend the contextual backdrop, four additional word clouds have been presented to exhibit the most commonly occurring bigrams associated with the candidates.

IV. RESULTS

After completing the data collection and cleaning processes, the presidential election data set contained a total of 119,276 tweets. With these tweets, the corpus, the vectorization and the classification models were created to classify the tweets with positive and negative sentiment toward each candidate.

In Figures 5 and 6, for Keiko Fujimori and Pedro Castillo, respectively, the F1-score results on the validation set for different configurations of ML models using the BOWE, TF-IDF and W2V vectorizers are shown. Both graphs display a similar pattern. Random Forest has higher and more consistent F1-scores, especially with TF-IDF and Word2Vec, with medians close to 0.9. Logistic regression ranks second, followed closely by Decision Trees. SVM, although having quite high scores, shows the greatest variability. Overall, the BOWE vectorizer does not perform as well as the other two vectorization approaches. The combination of Random Forest and W2V performs best overall. For Logistic Regression, TF-IDF shows better results than W2V. For the remaining cases, both W2V and TF-IDF exhibit similar performance.

Then, the top ten combinations for each candidate were selected. As shown in Tables 8 and 9 for Keiko Fujimori and Pedro Castillo respectively, it can be observed that in both cases the best vectorizer was W2V. For Keiko, the best model was SVM, while for Pedro it was RF. Furthermore, both the F1-score metric and the accuracy followed the same trend, achieving an accuracy greater than 0.93 for both candidates.

After identifying the most accurate model for each dataset, we predicted the sentiment conveyed in the tweets with

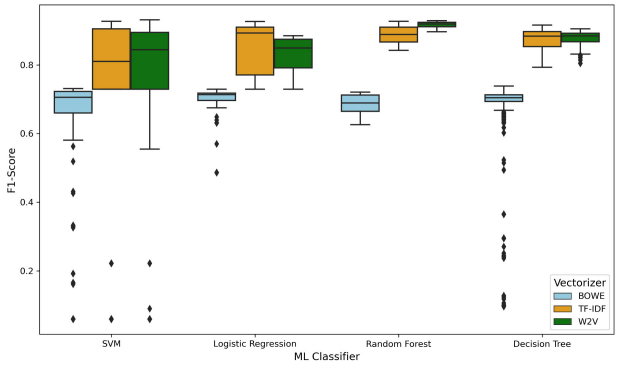


FIGURE 5. Box plot of F1-Scores for each classifier and vectorizer on evaluation data - Keiko Fujimori.

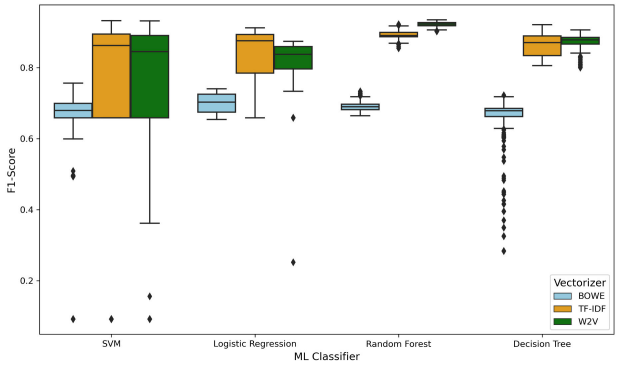


FIGURE 6. Box plot of F1-Scores for each classifier and vectorizer on evaluation data - Pedro Castillo.

TABLE 8. Top ten best model results for Keiko's dataset.

F1 Score	Accuracy	Model	Vectorizer
0.931607	0.934783	SVM	Word2Vec
0.928972	0.931884	RandomForest	Word2Vec
0.928283	0.931159	SVM	Word2Vec
0.928283	0.931159	SVM	Word2Vec
0.928149	0.931159	RandomForest	Word2Vec
0.928149	0.931159	RandomForest	Word2Vec
0.928013	0.931159	RandomForest	Word2Vec
0.927729	0.930435	RandomForest	Word2Vec
0.927693	0.929710	SVM	Word2Vec
0.927461	0.930435	RandomForest	Word2Vec

TABLE 9. Top ten best model results for Pedro's dataset.

F1 Score	Accuracy	Model	Vectorizer
0.934996	0.937255	RandomForest	Word2Vec
0.933920	0.936275	RandomForest	Word2Vec
0.932326	0.933333	SVM	TF-IDF
0.932326	0.933333	SVM	TF-IDF
0.931887	0.934314	SVM	Word2Vec
0.931635	0.934314	RandomForest	Word2Vec
0.931635	0.934314	RandomForest	Word2Vec
0.930807	0.933333	SVM	TF-IDF
0.930807	0.933333	RandomForest	Word2Vec
0.930679	0.933333	SVM	Word2Vec

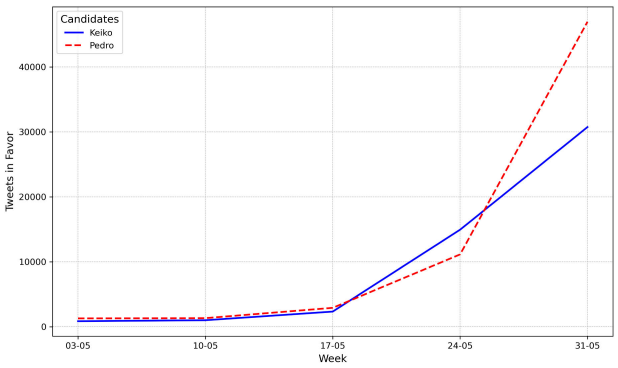


FIGURE 7. Comparative weekly favorability in tweets for candidates.

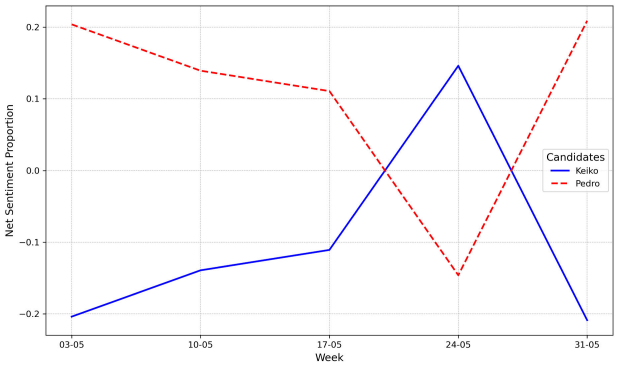


FIGURE 8. Net sentiment proportion for candidates by week.

the ML sentiment classification model. On the basis of these predictions, we constructed two figures. Figure 7 presents a visual representation of the cumulative positive sentiment associated with each candidate. This metric is derived by adding positive posts related to a candidate and negative posts related to their adversary, providing a complete sentiment profile. In contrast, Figure 8 delineates the net weekly sentiment for each candidate. This is calculated as the difference between the proportion of positive tweets and the proportion of negative tweets about a candidate, also evaluated on a weekly basis.

These visualizations are pivotal as they encapsulate the essence of the electoral mood, revealing how public sentiment fluctuated in response to the candidates' campaigns and potentially impacted the election results. Figure 7 demonstrates Pedro Castillo's lead in positive tweet volume, reflecting the actual election outcome where he won the presidency of Peru.

In Figure 8, an interesting behavior is described in the Keiko Fujimori data, where she shows a favorable trend during the week of the debates, which, however, diminishes in the following week.

Finally, four-word clouds were generated to underscore the most pertinent terms associated with positive and negative sentiments toward the two presidential contenders. These word clouds were derived from the training data set that was manually annotated. In this sense, Figures 9 a and b illustrate the frequently recurring words in positive and negative social media posts regarding Pedro Castillo, respectively. Similarly, c and d represent the predominant words in positive and negative tweets about the alternate presidential candidate, Keiko Fujimori. On a more detailed examination of the sentiment manifested in the evaluated posts, it is pertinent to highlight the predominance of negative commentary, which outnumbers the positive remarks by a factor of three.

Regarding Pedro Castillo, the favorable tweets predominantly contained affirmations of support, exemplified by phrases such as "come on" or "good". It should be noted that certain aspects of his policy proposals have received a positive reception among the electorate (such as health). Furthermore, affirmative comments were frequently associated with media outlets, including America, Gestión, ATV (Juliana Oxenford), RPP, and Wayka. In contrast, the critical posts largely pertained to his technical team and his envisioned strategies for the nation's future. In particular, his association with Vladimir Cerrón elicited considerable disquietude among the populace. Concerns were also raised about Castillo's stance on economic and constitutional issues. In social media, it is commonplace for public figures to be subjected to satire; in Castillo's case, this involved comparisons with the comedic character Cantinflas.

In the context of Keiko Fujimori, the tweets that conveyed approval similarly employed general affirmations of support such as "come on" or "good". These tweets often referred to her technical team and policy proposals. The political perspective in these tweets was characteristically optimistic, with particular emphasis on the themes of the economy and democracy, which were perceived as strengths in her campaign. The media outlets associated with positive coverage of Fujimori included Willax, ATV (Juliana Oxenford), Exitosa, RPP, and America. In contrast, the predominant themes in the negative tweets were linked to allegations of corruption and accusations of populism, the latter evidenced by the frequent mention of excessive promises of aid.

Additionally, recurrent, yet tangential, topics such as her constant mention of figure Vladimir Cerrón and an unsavory incident involving an assault on a congressman from her party were met with significant disapproval from the electorate. Compounding the negative sentiment were references to the actions of her father during his presidency, including the contentious policies of forced sterilizations and purported torture involving her mother. These historical references contributed to a substantial negative perception of her candidacy.

For a more quantitative analysis, Tables 10 (Keiko) and 11 (Pedro) display the top 10 most frequent words for each candidate, both positive and negative. In both cases, it is evident that tweets referring to both candidates frequently include words strongly associated with positive sentiments (e.g., "come on", "good", "health") and negative sentiments (e.g., "Cerron", "proposal(s)"). This indicates clear support or opposition from the electorate.



a. Positive words - Pedro Castillo

b. Negative words - Pedro Castillo

c. Positive words - Keiko Fujimori

d. Negative words - Keiko Fujimori

FIGURE 9. Most frequent words related to each candidate.

TABLE 10. Positive and negative words in tweets regarding Keiko Fujimori.

Positive Words	Frequency	Negative Words	Frequency
Come on	60	Corruption	114
Peru	54	Peru	112
Good	47	Cerron	94
Health	41	Stone	93
DecideWellWillax	41	Bond	92
Proposals	39	Women	84
Team	37	Government	75
Technical	36	Bonds	72
Better	35	Fujimorism	65
Government	31	Proposals	63

TABLE 11. Positive and negative words in tweets regarding Pedro Castillo.

Positive Words	Frequency	Negative Words	Frequency
Come on	46	Proposals	107
Health	44	Cerron	94
Good	28	Proposal	66
Corruption	20	Team	66
Talk	20	Technical	65
Segment	20	Like	63
PeruLibrePress	19	Country	52
Elections	19	Plan	48
Country	18	Datum	46
Datum	17	So	44

Moreover, to dive deeper into the nuances of sentiment expressed in the tweets, it becomes imperative to contextualize the individual words within meaningful pairs to adequately understand the context in which words are used to describe candidates in a positive or negative light. Recognizing this necessity, our study extends its analysis beyond the single-word examination to explore bigrams—pairs of consecutive words that offer a more nuanced understanding of sentiment than the single-word analysis. This methodological expansion allows us to uncover intricate language patterns and contextual cues related to each candidate, providing richer insight into the sentiments expressed by Twitter/X users. The insights gleaned from this bigram analysis are presented in Figure 10, which reveals intricate language patterns related to each candidate, offering a comprehensive perspective on the discourse surrounding the presidential contenders and their respective campaigns.

Regarding the analysis of bigrams, the findings for Pedro Castillo reveal that concerning positive sentiment, there was a public endorsement for transformation, predicated on the candidate's assurances of deviating from the practices of previous politicians and his commitment to combat criminal organizations entwined with government officials. In contrast, there was palpable concern among the population about his perceived lack of capability, the limited expertise of his technical team, his connections with Vladimir Cerrón, and his position on gender-related policies.

In the case of Keiko Fujimori, the bigram analysis indicates a significant prominence of the Willax media outlet in association with positive remarks, as well as references to her technical team, tourism, and the oxygen bond, the latter about the pandemic context. With regard to negative comments, the analysis provided more comprehensive information. The recurrent mention of Vladimir Cerrón, a non-direct adversary, and the controversial governance of



a. Positive bigrams - Pedro Castillo

b. Negative bigrams - Pedro Castillo

c. Positive bigrams - Keiko Fujimori

d. Negative bigrams - Keiko Fujimori

FIGURE 10. Most frequent bigrams related to each candidate.

her father were both notable themes, echoing the findings from the individual word analysis. In addition, the results highlighted concerns about her lack of political preparedness and the perceived gaps in the information she presented during the presidential debate.

Similarly, the quantitative data for bigrams is shown in Tables 12 (Keiko Fujimori) and 13 (Pedro Castillo). In this case, the frequency is much lower than that of individual words. However, it provides a clearer view of the overall landscape, as it is more evident that favorable tweets for Keiko are related to the media outlet Willax, and the meaning of certain expressions like "clean hands" in favor of Pedro is better understood.

The results of our sentiment analysis provide valuable insight into the dynamics of public opinion surrounding the Peruvian presidential elections of 2021, shedding light on trends that closely mirror those observed in published polls. However, our analysis also unveils nuanced patterns and limitations that merit further exploration. In this discussion section, we dive into these findings, exploring the factors driving shifts in sentiment toward each candidate, and addressing the implications of our methodology choices. Additionally, we examine the limitations of our study and offer information on future research directions.

TABLE 12. Positive and negative bigrams associated with Keiko Fujimori.

Positive Bigrams	Frequency
WillaxTv DecideWellWillax	26
Technical Team	15
DecideWellWillax DebateInWillax	9
RppNews PowerInYourHands	8
Survey Datum	8
KeikoPresident UnitTheUniqueOption	7
PoliticsPe PoliticasInWillax	7
PoliticasInWillax DecideWellWillax	7
decidebienwillax KeikoFujimori	7
Bond Oxygen	6
Negative Bigrams	
Cerron Cerron	43
Forced Sterilizations	41
Peru Possible	28
Thousand Soles	26
MemoriesPeru KeikoFujimori	25
Every Time	20
Fight Corruption	20
I Dont Think So	20
Can Not	19
If Not	18

V. DISCUSSION

In this paper, we evaluate the performance of several Machine Learning algorithms in combination with a diversity of vectorizers to detect the sentiment (positive or negative) contained in tweets regarding the candidacies of Keiko Fujimori and Pedro Castillo in the context of the Peruvian presidential elections in 2021. Despite the attempts of the scientific community to employ sentiment analysis in electoral contexts around the world, there seems to be no common criteria regarding the choice of vectorizer and

TABLE 13. Positive and negative bigrams associated with Pedro Castillo.

Positive Bigrams	Frequency
Can Not	12
We Won't	10
Criminal Organization	9
Tia Maria	9
Environment	9
Survey Iep	8
Clean Hands	8
Technical Team	8
Technical Peru	8
Electrocute Mother	8
Negative Bigrams	
Peru No	44
Technical Team	40
Don't now	28
Lady Lady	28
Gender Approach	23
Vladimir Cerron	22
Peru Posible	21
Word Teacher	21
Technical Peru	19
Government Plan	19

classifiers to tackle this kind of problem. However, it is still possible to find some commonalities with the existing literature.

Regarding vectorizers, our results reveal that, regardless of the classifier, TF-IDF and W2V are the vectorizers with the highest impact on tweet representation, which aligns with the results obtained by [7] and [24]. However, we should mention that, in our case, W2V exhibits a small superiority in terms of accuracy and robustness (according to the F1-Score), as can be seen in Figures 5 and 6. On the other hand, our results show that the best classifiers for tweet sentiment analysis are between SVM and RF, being that SVM obtained better results for the case of Keiko Fujimori and RF stood out slightly for the case of Pedro Castillo; these results are compatible with the results exposed by [18] and [22], as shown in Tables 8 and 9, respectively.

Furthermore, as seen in the results, sentiment analysis proves to be a valuable tool in predicting electoral outcomes between certain candidates, with trends closely mirroring those observed in published polls. However, our analysis unveils nuanced patterns and limitations that merit further exploration.

An interesting trend revealed by our analysis is the increase in favorable sentiment toward Keiko Fujimori during a specific period. This increase may be attributed to various factors, including the increased engagement of Fujimori supporters on social media compared to those of Pedro Castillo, especially on Twitter/X. It should be noted that our sentiment classification process, relying on manual labeling for 10% of the tweets, introduces a degree of bias that may have influenced the results. The bias of the human labeler is mitigated by using sentiment analysis software and text analytics tools that apply natural language processing techniques to analyze emotional responses, decipher the true meaning of comments and generate consensus from multiple labelers [28].

In this study, the W2V and TF-IDF vectorizers demonstrated similar performance. W2V learns from the context in which words are used; however, with small datasets, it cannot properly capture the semantic and syntactic information of the words [24]. However, despite its lower ability to capture semantic and syntactic information in small data such as tweets, TF-IDF proved to be highly accurate as a feature extractor.

These findings highlight the importance of adapting the methodology to the unique characteristics of the data at hand. However, limitations such as difficulties in detecting sarcasm and the presence of repeated tweets by the same user underscore the need for further refinement in future studies, especially when processing tweets containing sarcasm, mockery, suggestions, among others, that make the labeling process complex (manually and by computer).

To address these challenges, pre-trained deep learning models have been employed. These models are trained in specific tasks related to sarcasm detection using labeled datasets, enabling them to identify subtle textual characteristics indicative of sarcasm, thus improving the accuracy of sentiment analysis in the presence of sarcastic and mockery comments [29], [30]. However, this particular aspect falls outside the scope of our current study and is reserved for future work.

Additionally, to emulate the one-person-one-vote principle, we limited our analysis to one tweet per user per week. Our approach, guided by the recommendations of Singh et al. [5], involved the systematic removal of duplicate tweets, retaining the first instance of each user weekly. However, this method may not fully encapsulate the breadth of a user's opinion, as it considers only a single tweet while potentially omitting subsequent tweets that could provide a more comprehensive view of their sentiment toward the candidates.

By comparing our findings with previous studies, both within Peru and around the world, we find alignment with the existing literature on the effectiveness of sentiment analysis in predicting electoral outcomes. However, our study offers unique insights into the Peruvian political landscape, shedding light on the distinctive patterns of public sentiment and engagement on social media platforms such as Twitter/X.

Looking ahead, future research could explore more advanced natural language processing techniques to overcome the limitations encountered in this study. Furthermore, considering the implications of our findings for election analysis, there is potential for further research on how insights from social media sentiment analysis can inform political strategies, media coverage, and voter engagement efforts in Peru and beyond.

In conclusion, while our analysis provides valuable insight into the dynamics of public opinion during the Peruvian presidential elections of 2021, it also underscores the need for ongoing refinement and innovation in sentiment analysis

methodologies. By addressing the challenges and taking advantage of the opportunities presented by social media data, we can continue to improve our understanding of political discourse and decision-making processes in the current digital age.

VI. CONCLUSION

In conclusion, this study has demonstrated the efficacy of sentiment analysis of Twitter/X data in predicting the results of the 2021 presidential elections in Peru. By systematically evaluating various feature extraction models, including TF-IDF, W2V, and BOWE, alongside Machine Learning techniques such as SVM, Random Forest, Decision Tree, and Logistic Regression, our analysis closely aligned with the 2021 electoral results. The results of the experiments showed that among the vectorizers, W2V and TF-IDF produced positive results, demonstrating their effectiveness in capturing meaningful features from the data. In contrast, BOWE lagged behind, indicating a need for further refinement. In terms of the best combinations, SVM with W2V (for Keiko's dataset) and Random Forest with W2V (for Pedro's dataset) proved to be the most effective, achieving F1-scores and accuracy metrics exceeding 0.93 for both candidates. This underscores the potential to accurately predict elections using social media analytics, leveraging advanced text mining and machine learning methods to reflect public sentiment toward candidates.

In this sense, our findings shed light on the salient issues of concern to the electorate, discernible through the discourse on social media. This highlights the instrumental role of social media as a platform for political candidates, not merely as a means of influence but also as a strategic tool to gauge public sentiment and tailor their campaign strategies accordingly, in response to the public's concerns. However, while single-word analysis provided valuable insight, the inherent limitations of this approach underscore the importance of considering context, as demonstrated by the utilization of bigram analysis to partially capture this aspect, since ideas are typically conveyed through combinations of words rather than isolated terms, consequently contributing to an area that is ready for a more in-depth exploration in subsequent researches.

In the future, it is crucial to optimize the labeling process and establish standardized criteria to minimize bias and enhance the accuracy of sentiment classification techniques. Additionally, the challenge of detecting sarcasm in tweets remains significant, impacting the precision of sentiment classification. Future research should also explore the feasibility of obtaining data by department to improve sample representativeness and correlation with official election results.

Finally, this study contributes to the growing body of literature on sentiment analysis and electoral prediction, providing valuable information for political campaigns and strategic decision making. By addressing methodological challenges and advancing our understanding of public sentiment dynamics, future research can build on these findings to further refine sentiment analysis methodologies and enhance their applicability in diverse political contexts.

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